Julian Bushlow

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Education

Cornell University Ithaca, NY

B.S. Electrical & Computer Engineering | GPA: 4.05, Major GPA: 4.14 *Aug. 2023 – May 2027*

Relevant Coursework: Digital ASIC, Computer Architecture, Deep Learning, Information Theory, Signals & Systems, Probability, Embedded Systems, Circuits, EM Waves, AI Hardware, Networks

Research & Projects

Undergraduate Researcher - Zhang Research Group, Cornell May 2026 – Present

Allo-Assisted ASIC Harness *Ithaca, NY*

- Extended the Allo dataflow architecture design framework with a hierarchical RTL → GDS ASIC backend using commercial EDA tools
- Modified the Allo MLIR dialect to preserve spatial and functional information for each processing element, enabling repeated structures to be identified and implemented hierarchically
- Developed a macro-based physical design flow that hardens representative subdesigns once and reuses them across large accelerator arrays, avoiding redundant synthesis and place-and-route work
- Characterized runtime, memory usage, and PPA agreement of hierarchical versus flat implementations across a range of accelerator designs; publication in preparation

Undergraduate Researcher - Abdelfattah Lab, Cornell Tech May 2025 – Present

FPGA Architectures to Enable Concurrent LUT and Adder Chain Usage *New York, NY*

- Co-authored an ACM TRETS paper introducing an FPGA logic block architecture that enables independent and concurrent LUT and adder-chain use within a single ALM, reducing area by 8–22% across three benchmark suites
- Designed and evaluated input-sharing architectures for ALM bypass paths to reduce added crossbar area while preserving logic packing efficiency
- Modeled Stratix-10-like ALM variants in VTR/VPR and COFFE, evaluating their area, timing, and packing behavior

Signal Processing Using FPGAs & TCD for Automated Emboli Detection

- Launched project with Weill Cornell Medicine to develop a compact headset using transcranial Doppler ultrasound methods and deep learning to detect stroke-causing microemboli with minimal doctor intervention
- Designed and implemented a PyMUST-based signal processing pipeline to transform and filter raw ultrasound echoes into inputs for deep-learning-based emboli detection
- Built ultrasound echo simulator using PyMUST for a fully parameterized blood vessel with realistic fluid dynamics and approximations of how flow velocities change over the cardiac cycle

Parameterized SNN Accelerator & ASIC Generator Spring 2026

ECE 6745 Final Project

- Co-designed a hardware-efficient Leaky Integrate-and-Fire neuron model, replacing multiplicative leakage with a shift-and-subtract approximation for integer-only digital implementation
- Built a parameterized generator for arbitrary spiking neural network topologies that emits RTL networks, memory-interface controllers, software wrappers, and randomized testbenches
- Developed a hierarchical RTL → GDS flow that hardens repeated submodules as reusable macros and automatically tiles them to reduce synthesis and place-and-route runtime
- Trained and deployed an N-MNIST classifier using hardware-aware fine-tuning for integer weight quantization and approximate neuron dynamics
- Benchmarked accelerator speedup, ASIC area, and tool runtime across network widths, depths, buffer sizes, and macro organizations; taped out a 2-layer, 8-neuron design in TSMC 180 nm

Publications

X. Dai, J. Pun, A. Y. Fei, **J. Bushlow**, G. Zgheib, M. A. Iyer, M. Eldafrawy, A. Boutros, V. Betz, and M. S. Abdelfattah, "FPGA Architectures to Enable Concurrent LUT and Adder Chain Usage," *ACM Transactions on Reconfigurable Technology and Systems (TRETS)*, 2026. Just Accepted. doi:10.1145/3834569

Teaching

ECE 3100 - Introduction to Probability | *TA* Jan. 2026 – May 2026

- Independently taught a weekly discussion section on probabilistic modeling, set theory, and problem-solving techniques, reviewing course material and introducing new concepts
- Wrote weekly reference solutions for problem sets, held office hours, graded homework, and proctored exams

ECE 2100 - Introduction to Circuits | *TA* Jan. 2025 – May 2025

- Ran biweekly labs, taught students to use lab equipment (including oscilloscopes, function generators, DC power sources, SMUs) and safely build circuits
- Prepared and taught material (DC & AC circuits, op-amp circuits, Laplace transforms & linear system theory) in discussion sections and office hours

Oral/Poster Presentations

AAAH - Allo-Assisted ASIC Harness: Oral presentation of research funded by Cornell's Office of Inclusive Excellence. Talk given at Summer 2026 OIE Symposium (slides)

FPGA-Based TCD Headset for Emboli Detection: Poster presentation of research funded by Cornell's Office of Inclusive Excellence. Delivered at Fall 2025 OIE Symposium (poster)

SpikeGPT Reimplementation: Cornell CS 4782 Deep Learning final project to reimplement a paper and present results. Delivered at CS 4782 Final Project Symposium (poster)

- Trained a spiking transformer and tested it on natural language understanding & generation (original paper)
- Best Project/Presentation winner (as voted by course staff)

Technical Skills

Languages/HDLs: Python, C/C++, SystemVerilog, Verilog, Bluespec SystemVerilog, RISC-V/ARM Assembly
EDA/Architecture Tools: Cadence Innovus, Synopsys Design Compiler, VCS, Calibre, Yosys, OpenROAD, VTR (VPR & COFFE), Quartus Prime, mflowgen
Frameworks/Libraries: PyTorch, snnTorch, PyMTL, NumPy, SciPy

Awards/Honors

Tau Beta Pi, Cornell Chapter (VP): Engineering honor society; top 12.5% of Cornell Engineering by GPA
Eta Kappa Nu, Cornell Chapter: IEEE national honor society, provides tutoring for ECE classes

Extracurricular Activities

Cornell Quizbowl | *President* May 2024 – May 2026

- Direct regional tournaments (15+ teams), organize biweekly practices, write question sets
- Plan cross-country trips to competitions, manage $5000 budget for a club of 100+ members